

No.	Solution	Remarks
1(a)(i)	The <u>gradient of the graph represents the acceleration</u>, which is <u>changing in value</u>. Thus <u>resultant force is changing</u>. This suggested the presence of air resistance, since <u>weight cannot be the only force</u> acting on the ball.	[1] [1]
1(a)(ii)	The magnitude of acceleration due to free fall can be determined from the <u>gradient of the graph at 1.75 s</u>. At this time, <u>velocity is zero which means there is no air resistance and the only force (net force) experienced by the ball is its own weight</u>.	[1] [1]
1(b)	At $t = 0$ s, the ball is moving with the largest speed and hence experiences the greatest air resistance. Gradient of tangent at $t = 0$ s, $= \frac{25}{1.2}$ $= 20.8$ Since $W + F_{air} = ma$, $(0.010 \times 9.81) + F_{air} = 0.010 \times \frac{25}{1.2}$ $F_{air} \approx 0.110 \text{ N } (\pm 10\%)$	[1] expl [1] for gradient at $t = 0$ s [1] ans
2(a)	Work done is area under the F - d graph.	[1]